1 Important Inequalities

The proofs of the error bound formulas all use the *Triangle Inequality*

$$\left|\sum_{i=1}^{n} x_i\right| \leq \sum_{i=1}^{n} |x_i|,$$

which you proved in AEP #1, as well as the following similar-looking inequality for integrals.

Proposition 1.1 (Triangle inequality for integrals)**.** *If $f : [a,b] \to \mathbb{R}$ is integrable then so is $|f|$ and*

$$\left|\int_a^b f(x)dx\right| \leq \int_a^b |f(x)|dx.$$

1. Assume the first statement in Proposition 1.1 and prove the second.
(*Hint*: $-|f(x)| \leq f(x) \leq |f(x)|$.)

2 The Midpoint Rule

Theorem 2.1 (Error bound for the midpoint approximations)**.** *Let $E_n^M(f)$ denote the absolute error $|\sum_{i=1}^n f(x_i^*)\Delta x - \int_a^b f(x)dx|$ for the the midpoint approximation. If $M = \max\{|f''(x)|\}_{x\in[a,b]}$, then $E_n^M(f) \le M\frac{(b-a)^3}{24n^2}$.*

To prove this theorem, we will need the following lemma.

Lemma 2.2.

$$f(b)-f(a) = \int_a^b f'(x)dx = \int_0^1 f'(a+t(b-a))(b-a)dt.$$

(a) Prove Lemma 2.2. (*Hint*: Use substitution.)

We now prove Theorem 2.1.

Proof. (b) For the midpoint approximation, show that the error can be written as

$$E_n^M(f) = \left|\sum_{i=1}^n L_i\right|,$$

where

$$L_i := \int_{x_{i-1}}^{x_i} f(x)dx - f\left(\frac{x_{i-1}+x_i}{2}\right)(x_i - x_{i-1}). \tag{2.3}$$

(c) Use Lemma 2.2 to show that Equation (2.3) can be written as

$$L_i = \int_{x_{i-1}}^{x_i} \left[\int_0^1 f' \left(\frac{x_{i-1}+x_i}{2} + t\left(x - \frac{x_{i-1}+x_i}{2} \right) \right) dt \right] \left(x - \frac{x_{i-1}+x_i}{2} \right) dx. \qquad (2.4)$$

(*Hint*: $\left(x - \frac{x_{i-1}+x_i}{2} \right)$ is independent of the integration variable t.)

(d) Integrate Equation (2.4) by parts by taking

$$u = \int_0^1 f'\left(\frac{x_{i-1}+x_i}{2} + t\left(x - \frac{x_{i-1}+x_i}{2}\right)\right) dt,$$
$$dv = \left(x - \frac{x_{i-1}+x_i}{2}\right) dt,$$

to show that

$$L_i = -\int_{x_{i-1}}^{x_i} \frac{d}{dx}\left[\int_0^1 f'\left(\frac{x_{i-1}+x_i}{2} + t\left(x - \frac{x_{i-1}+x_i}{2}\right)\right) dt\right] \frac{1}{2}(x-x_{i-1})(x-x_i). \qquad (2.5)$$

(*Hint:* Note that we can take v to be any antiderivative of $\left(x - \frac{x_{i-1}+x_i}{2}\right)$. Things work out most nicely by taking $v = \frac{1}{2}(x-x_{i-1})(x-x_i)$.)

(e) Assuming that you can move the derivative under the integral sign (after taking MVC you will be able prove this step is valid here); i.e.,

$$\frac{d}{dx}\int_0^1 f'\left(\frac{x_{i-1}+x_i}{2}+t\left(x-\frac{x_{i-1}+x_i}{2}\right)\right)dt=\int_0^1 \frac{d}{dx}f'\left(\frac{x_{i-1}+x_i}{2}+t\left(x-\frac{x_{i-1}+x_i}{2}\right)\right)dt,$$

show that Equation (2.5) becomes

$$L_i=-\int_{x_{i-1}}^{x_i}\left[\int_0^1 f''\left(\frac{x_{i-1}+x_i}{2}+t\left(x-\frac{x_{i-1}+x_i}{2}\right)\right)tdt\right]\frac{1}{2}(x-x_{i-1})(x-x_i)dx. \qquad (2.6)$$

(f) Now assume $M = \max\{|f''(x)|\}_{x\in[a,b]}$ and substitute Equation (2.6) into $E_n^M(f) = |\sum_{i=1}^n L_i| \leq \sum_{i=1}^n |L_i|$ to derive the bound

$$E_n^M(f) \leq M\frac{(b-a)^3}{24n^2}.$$

□